

ABANGCEBU AI — SPRINT 1 UI/UX ARCHITECTURE SPECIFICATION

Profile Page & Account Settings Wireframe Specification

3-Tab Zero-Trust Architecture (Personal Info, Security & Auth, KYC Verification) · MacBook Air 16:10 Frame (1440px)

Jira Reference: SCRUM-68

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Design Standards: Balsamiq/Figma Grayscale Blueprint Conventions, Apple HIG Ergonomics, Supabase RLS Boundary Isolation, WCAG 2.2 AA Contrast & Touch Rules

Framework: Next.js 16 + React 19 + Supabase RLS

Reviewed By: Hermar Centillas (Lead / Scrum Master)

Fidelity: MID-FIDELITY WIREFRAME

Status: APPROVED / ACTIVE

1. Executive Summary & Zero-Trust Identity Architecture

The Profile Page & Account Settings interface (/profile) establishes the operational trust foundation for AbangCebu AI. In the Metro Cebu boarding house ecosystem, where unverified listings and fraudulent advance deposits have historically victimized student renters near universities (CIT-U, USC, UC, USJ-R) and professionals in Cebu IT Park, identity verification is essential.

Zero-Trust Security & Spatial Governance: AbangCebu AI enforces four architectural pillars: **(1) Anti-Scam KYC Trust Verification** with cryptographic validation of Philippine IDs (PhilSys, UMID, Driver's License, PRC ID); **(2) Spatial Corridor Preference Engine** driving algorithmic radar matches along university corridors (e.g. CIT-U on N. Bacalso Ave); **(3) Zero-Trust Session & NIST SP 800-63B Authentication** with 4-segment entropy gauge and 1-tap emergency session kill-switch; and **(4) Supabase Row-Level Security** guaranteeing private data isolation with AES-256 encrypted KYC storage.

2. Desktop Viewport Specification (1440px × 900px · MacBook Air Chassis)

Encased in an authentic MacBook Air chassis (16:10 display), desktop navigation is anchored by a 260px fixed dark sidebar (#0F172A) with *Profile & Settings* highlighted in Coastal Emerald (#047857). The 1180px main workspace features a profile header hero card, accessible 3-tab bar, and integrated personal details, security, and KYC containers:

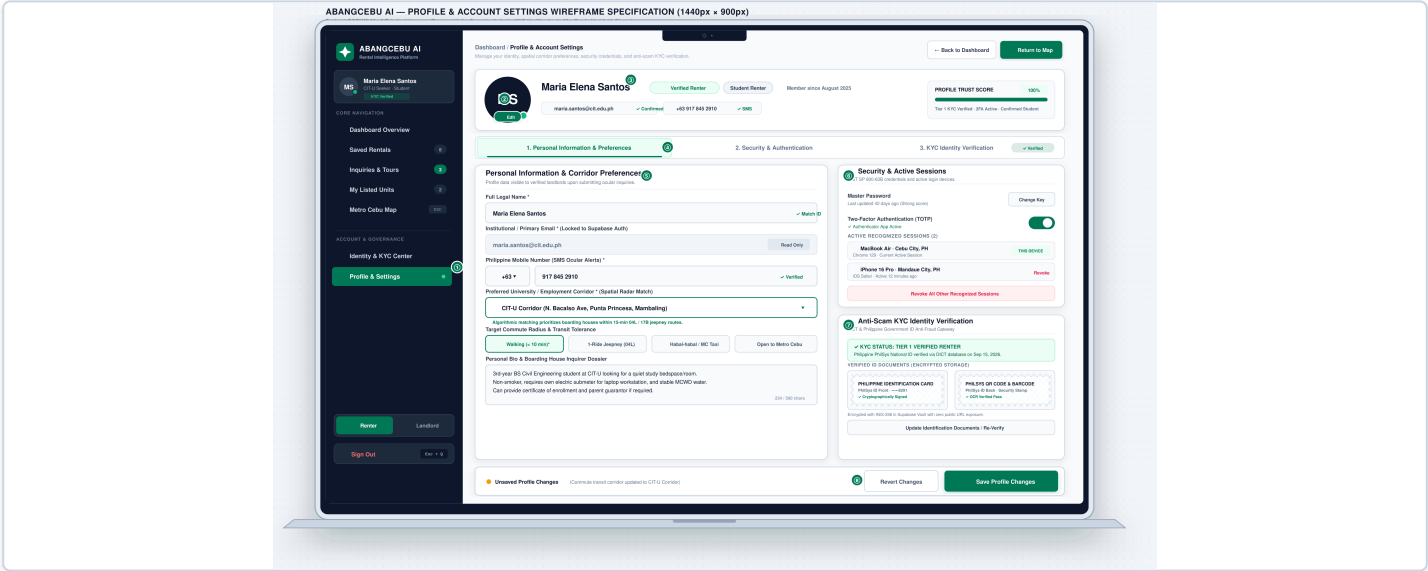


Figure 1: Desktop 1440px Mid-Fidelity Wireframe Blueprint — MacBook Air Aluminum Chassis, 260px Fixed Sidebar with Highlighted Profile Nav, Hero Card with 88px Avatar & Edit Trigger, 3-Tab Bar, Personal & Corridor Form, Security Sessions & Anti-Scam KYC Gateway

Component	Dimensions / Placement	Wireframe Role & Authentic Design Architecture
MacBook Air Chassis	1480 × 950px Outer	Aluminum unibody frame, top FaceTime camera notch, camera lens, green LED, inner bezel, and bottom opening lip.
Fixed Sidebar Nav ①	w: 260px, h: 900px (x: 0, y: 0)	Dark surface (#0F172A), brand crest, user capsule, nav items, Profile & Settings active in Emerald (#047857), and bottom Sign Out.
Profile Header Hero ②	w: 1120px, h: 116px (x: 284, y: 74)	Elevated card with 88px avatar, [📄 Edit] badge trigger, online dot, full name (Maria Elena Santos), and 100% Trust Completion gauge.
Trust & Contact Badges ③	h: 22px – 24px each	🛡 Verified Renter pill (#ECFDF5), Student tag, confirmed institutional email badge, and verified Philippine mobile badge (+63 SMS).
Accessible 3-Tab Bar ④	w: 1120px, h: 42px (x: 284, y: 202)	WAI-ARIA tablist (role="tablist") with Tab 1 Personal Info & Preferences (active), Tab 2 Security & Auth, and Tab 3 KYC Verification.
Personal & Corridor Form ⑤	w: 670px, h: 560px (x: 284, y: 256)	Inputs for Legal Name, locked Email, Philippine Mobile (+63), CIT-U Corridor dropdown, commute tolerance pills, and Bio textarea.
Security & Sessions Module ⑥	w: 430px, h: 270px (x: 974, y: 256)	Password status, 2FA TOTP toggle, active device sessions (MacBook Air + iPhone 16 Pro), and emergency 🚨 Revoke All Other Sessions CTA.
Anti-Scam KYC Gateway ⑦	w: 430px, h: 275px (x: 974, y: 541)	Tier 1 Verified status banner, encrypted storage PhilSys ID Front and Back blueprint containers, and document re-verification trigger.
Sticky Bottom Action Bar ⑧	w: 1120px, h: 54px (x: 284, y: 830)	Unsaved changes indicator, Revert Changes button, and primary 📁 Save Profile Changes button (44px height, #047857).

ABANGCEBU AI — MOBILE INTERACTION ARCHITECTURE

Mobile Viewport Wireframe Specification (393px × 852px · 100dvh)

Touch Ergonomics & 4 Operational States · iPhone 16 Pro Titanium Chassis · iOS Safari 100dvh

3. Mobile Viewport Geometry & Touch Ergonomics

The mobile profile interface standardizes on 100dvh (Dynamic Viewport Height) encased in an authentic iPhone 16 Pro Titanium chassis with Dynamic Island and persistent bottom iOS Safari navigation. It enforces 44×44px touch targets and the 16px input font rule. Below is the side-by-side comparative blueprint across 4 core operational states:

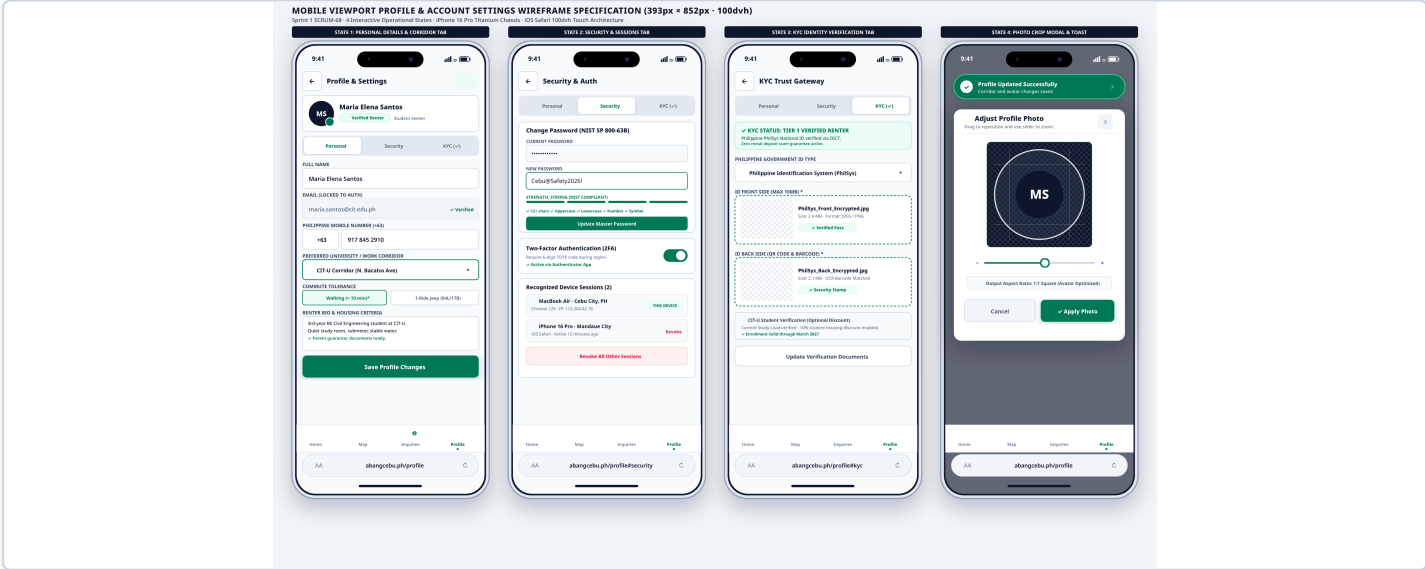


Figure 2: Mobile 100dvh Mid-Fidelity Wireframe Blueprint — iPhone 16 Pro Titanium Chassis, Phone 1 (Personal & Corridor Tab), Phone 2 (Security & NIST SP 800-63B Tab), Phone 3 (KYC Verification Tab), and Phone 4 (Avatar Crop Modal & Floating Success Toast)

Operational State	Visual Hierarchy & Components	User Interaction & Workflows	Ergonomic & Technical Governance
Phone 1: Personal Tab	Compact hero capsule (52px avatar, camera button, name, verified pill), segmented 3-tab bar, single-column vertical form canvas.	Edit full name, view locked institutional email, configure +63 mobile, select CIT-U university corridor, and edit bio dossier.	Strict 16px mobile input font rule; 44×44px touch targets; bottom tab bar with Profile active in Cebu Emerald.
Phone 2: Security Tab	Change password card, NIST SP 800-63B 4-segment entropy gauge, 5 requirement pills, 2FA toggle, active device sessions list.	Update master password with real-time strength feedback; toggle TOTP 2FA; triage recognized devices (MacBook Air, iPhone 16 Pro).	Emergency 1-tap [🚨 Revoke All Other Sessions] trigger; high-contrast security indicators; ARIA switch semantics.
Phone 3: KYC Gateway Tab	Tier 1 Verified status banner, Philippine government ID selector dropdown (PhilSys), dual front and back document containers.	Review verified identity status; select government ID type; tap-to-capture or drag-and-drop document upload (max 10MB).	Encrypted AES-256 private storage; zero public URL exposure; student enrollment verification for 10% rent discount.
Phone 4: Crop Modal & Toast	Dimmed modal backdrop (65% opacity), circular viewfinder cropping mask (85px radius), zoom slider, floating success toast.	Drag to reposition and slider to zoom photo; apply photo; auto-dismissing toast: "Profile updated successfully".	WAI-ARIA modal dialog trap; 1:1 aspect ratio constraint; role="status" polite toast announcement.

MOBILE TOUCH ERGONOMICS & ZERO-TRUST GOVERNANCE

- WCAG 2.2 AA 44×44px Touch Targets: Enforced via @utility touch-target on all buttons, tabs, avatar edit triggers, and toggle switches.
- Anti-Zoom 16px Rule: Mobile inputs, dropdown selectors, and textareas strictly locked to 16px (text - [16px]) to eliminate destructive iOS Safari zoom.
- Thumb Zone Ergonomics: Primary save and update buttons placed within lower two-thirds natural thumb reach.
- Anti-Scam KYC Trust Pipeline: Verified badges and direct phone access unlocked only upon valid government ID validation, eliminating deposit fraud.

Accessibility, Route Architecture & Sign-Off

WCAG 2.2 AA Sequential 16-Step Tab Order · Next.js 16 Server/Client Boundaries · Definition of Done

4. Sequential Keyboard Tab Order & WAI-ARIA Matrix

Tab #	Element	Accessible Name	ARIA Roles & States	Keyboard Action
1	Brand Crest Link	"AbangCebu AI Home"	role="link"	Enter routes to home
2	Profile Mini Capsule	"Maria Elena Santos, Student Seeker"	role="region", aria-label="Profile Dossier"	Visual anchor / Tab next
3	Profile & Settings Nav	"Profile & Account Settings, current page"	role="link", aria-current="page"	Current view indicator
4	Role Workspace Switch	"Switch Workspace Role"	role="tablist", aria-label="Role Switch"	ArrowRight/Left toggles role
5	Sign Out Button	"Sign Out of AbangCebu AI"	role="button", aria-label="Sign Out"	Enter opens logout dialog
6	Back to Dashboard CTA	"Return to Dashboard Overview"	role="link", aria-label="Back to Dashboard"	Enter routes to /dashboard
7	Return to Map CTA	"Return to Metro Cebu Map Explorer"	role="link", aria-label="Return to Map"	Enter routes to /map
8	Avatar Edit Button	"Change Profile Photo"	role="button", aria-haspopup="dialog"	Space/Enter opens crop modal
9	Tablist (3 Tabs)	"Profile Navigation Sections"	role="tablist", aria-label="Profile Tabs"	ArrowLeft/Right cycles tabs
10	Tab 1: Personal Info	"Personal Information and Preferences Tab"	role="tab", aria-selected="true"	Enter activates tab
11	Tab 2: Security & Auth	"Security and Authentication Tab"	role="tab", aria-selected="false"	Enter activates tab
12	Tab 3: KYC Verification	"KYC Identity Verification Tab, Verified"	role="tab", aria-selected="false"	Enter activates tab
13	Full Legal Name Input	"Full Legal Name"	type="text", aria-required="true"	Standard text input
14	Philippine Mobile Input	"Philippine Mobile Number (+63)"	type="tel", aria-required="true"	Numeric 10-digit input
15	Corridor Selector	"Preferred University or Employment Corridor"	role="combobox", aria-expanded="false"	ArrowDown opens menu
16	Save Profile Changes	"Save Profile Changes"	role="button", aria-label="Save changes"	Enter submits profile

5. Next.js 16 Component Hierarchy & Boundaries (Sprint 2 Contract)

Component	Runtime	File Location (Sprint 2)	Architectural Role & Contract
ProfilePage	SERVER	src/app/(dashboard)/profile/page.tsx	Enforces Supabase session auth. Prefetches profiles, auth.users, active sessions, and KYC document status.
ProfileHeader	SERVER	src/components/profile/profile-header.tsx	Hero card container. Renders 88px avatar, trust verification pills, contact badges, and profile trust completion gauge.
AvatarUploadTrigger	CLIENT	src/components/profile/avatar-upload-trigger.tsx	Handles avatar click, file input change, image preloading, and launches the AvatarCropModal.
ProfileTabContainer	CLIENT	src/components/profile/profile-tab-container.tsx	Manages active tab state (personal, security, kyc), URL hash synchronizer, and ARIA keyboard arrows.
PersonalDetailsTab	CLIENT	src/components/profile/personal-details-tab.tsx	Manages form state, dirty field tracking, corridor matching preference updates, and connects to the bottom action bar.
SecurityTab	CLIENT	src/components/profile/security-tab.tsx	Hosts NIST SP 800-63B password change form, 2FA TOTP toggle, and device session list with revoke mutations.
KycVerificationTab	CLIENT	src/components/profile/kyc-verification-tab.tsx	Handles drag-and-drop document uploads to encrypted Supabase bucket, OCR pre-validation, and displays 4 status banner tiers.
ActiveSessionsList	CLIENT	src/components/profile/active-sessions-list.tsx	Displays recognized active devices with IP addresses, browser agents, and 1-tap Revoke All Other Sessions handler.
AvatarCropModal	CLIENT	src/components/profile/avatar-crop-modal.tsx	Client dialog featuring circular viewfinder mask, canvas zoom slider, and 1:1 image blob generator.
StickyActionBar	CLIENT	src/components/profile/sticky-action-bar.tsx	Floating footer that detects form dirty state, provides Revert action, and executes optimistic profile updates.

6. Definition of Done & Architectural Sign-Off

- Mid-Fidelity Wireframe Standard: Balsamiq/Figma grayscale blueprints, crossed [X] placeholders, and authentic MacBook Air / iPhone 16 Pro chassis.

Desktop 3-Tab Specification: 1440×900px, 260px fixed dark sidebar, hero card with 88px avatar & edit badge, trust badges, personal form, security snapshot, KYC gateway, and sticky action bar.

Mobile 4-State Specification: 100dvh viewport, Phone 1 (Personal Tab), Phone 2 (Security Tab), Phone 3 (KYC Tab), and Phone 4 (Avatar Crop Modal & Floating Toast).

Anti-Scam KYC Trust Gateway: 4 deterministic tiers (unverified, pending, verified, rejected) and encrypted document containers.

Accessibility Compliance: WCAG 2.2 AA 44×44px touch targets, 16px mobile input anti-zoom rule, contrast >= 4.5:1, and sequential 16-step tab order flow.

Sprint 1 Boundary Maintained: Purely architectural wireframe blueprints and data contracts; ZERO premature React/JSX feature code.

Rendered Engineering Assets: High-res vector SVGs, 200 DPI PNG blueprints, and 3-page WeasyPrint engineering specification PDF.

ARCHITECTURAL SIGN-OFF

Author: Angel Crushein Yaun (UI/UX Designer)
Reviewer: Hermar Centillas (Lead / Scrum Master)
Jira Ticket: SCRUM-68 | Status: Approved & Audited
Date: September 30, 2026